Trained PING streams sequential digits without retraining

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Abstract

A PING network trained one digit at a time on 200 ms trials can classify a *stream* of digits at a fraction of that time each, with no retraining: only the readout's integration window changes. At τ = 50 ms (≈ 2 gamma cycles) 4 of the 5 sequential MNIST digits classify correctly, the readout flipping within one cycle of each transition; a 2-D sweep over (τ, input rate) reveals a sub-cycle failure floor below ≈ 15 ms and a broad high-accuracy plateau above it, where short presentation time and weak drive trade off along iso-accuracy diagonals.

Method

Canonical exp025 PING baselines (1,024 excitatory + 256 inhibitory cells, trained one MNIST digit per 200 ms trial, three seeds 42, 43, 44). Everything here is **inference-only** at the trained Δt = 0.10 ms; the weights are never updated. The 2-D sweep averages over the three seeds; the single-stream demos use seed 42.

A stream of digits, each shown for τ ms, is classified in one forward pass. The *only* change from training is a sliding readout window:

1. **Encode** each digit as a Poisson spike train over τ ms across 784 input channels, one per pixel, firing rate proportional to pixel intensity at the chosen input rate.
2. **Concatenate** the per-digit trains into one input stream. Transitions are *hard switches*: the Poisson rate flips instantaneously at $t = k\tau$, with no blending.
3. **One forward pass** of the trained network over the whole stream (no retraining), at the trained Δt.
4. **Integrate evidence** in a non-spiking output leaky-integrator, one unit per class:

$$v_{\text{out}}(t) = \beta_{\text{out}} v_{\text{out}}(t-1) + \frac{1-\beta_{\text{out}}}{\Delta t} \boldsymbol{s}^E(t-1) W_{\text{out}}.$$

5. **Read a sliding window.** Average v_{out} over the trailing τ-window and softmax it:

$$\text{logits}(t) = \frac{\Delta t}{\tau} \sum_{u=t-w+1}^{t} v_{\text{out}}(u), \quad p(\text{class}, t) = \text{softmax}(\text{logits}(t)),$$

   At training the average ran over the *whole* trial; the trailing window is the single change.

6. **Predict per segment** as $\arg\max_c p(\text{class} = c, t)$, read at the end of each digit's τ-window.

Here $\boldsymbol{s}^E(t)$ is the excitatory spike vector, W_{out} the trained readout (unchanged), and $\beta_{\text{out}} = \exp(-\Delta t/\tau_{\text{out}})$ the output leak with $\tau_{\text{out}} = 2$ ms. The probability trace $p(\text{class}, t)$, the network's online confidence in each digit, is what the figures plot.

The grid sweeps τ over 10, 15, 25, 40, 50, 75, 100, 200 ms and input rate over 5, 10, 25, 50, 100, 200 Hz per channel, giving 8 × 6 cells, each 40 streams × 10 digits × 3 seeds (1200 segments per cell). It extends down to τ = 10 ms (≈ 0.36 of a gamma cycle) to resolve the sub-cycle regime.

Results

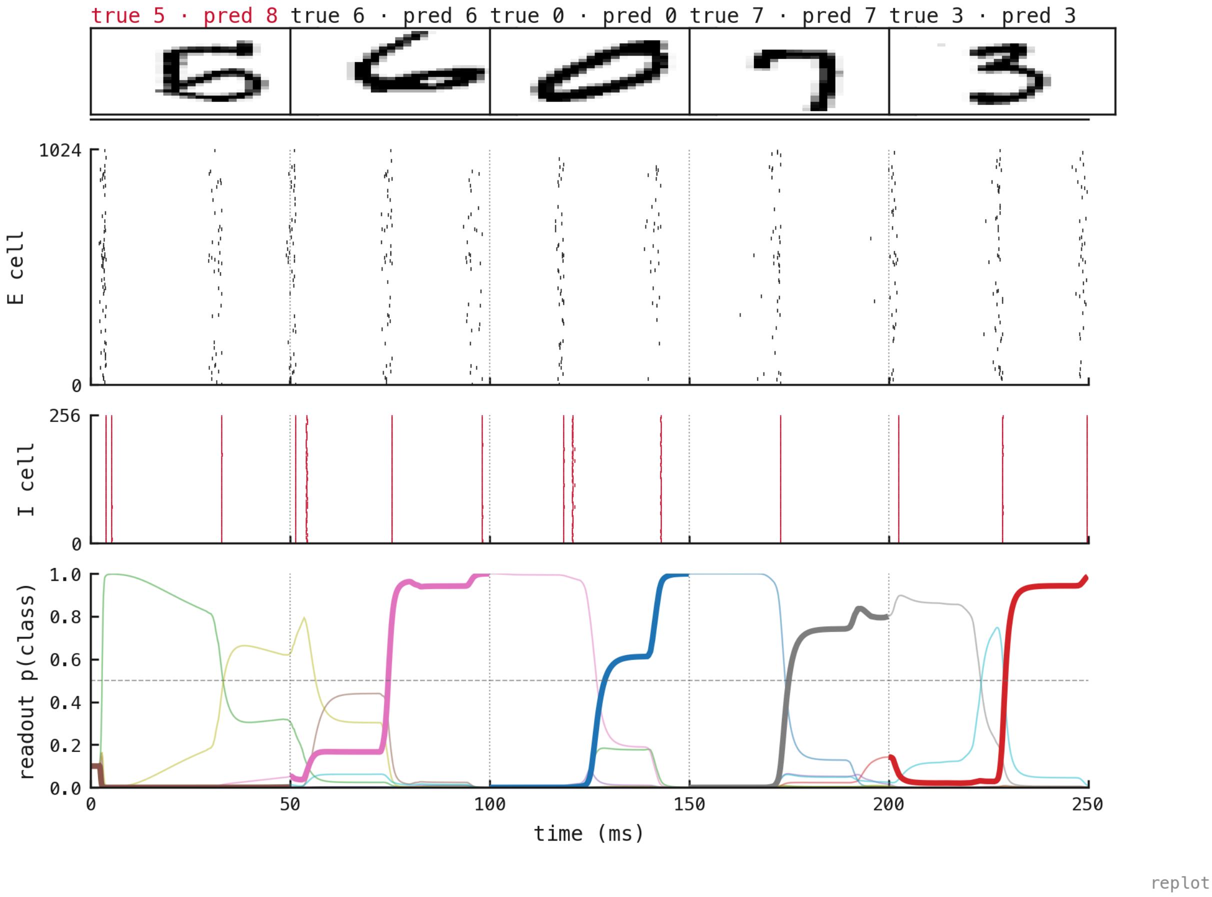

Figure 1: Five sequential digits (5, 6, 0, 7, 3) at $\tau = 50$ ms each ($\approx$ 2 gamma cycles per digit, 250 ms total, 25 Hz input). The gamma cadence ($\approx$ 28 ms) is preserved across the stream; the readout flips to the new digit within one cycle of each transition and reaches near 1.0 by the segment's end. **4/5 correct** (the leading 5 is misread). The figure makes four things visible at once: sparse E firing, the I-burst clock, class-tracking readout, and re-identification within $\approx$ 1 cycle of each switch.

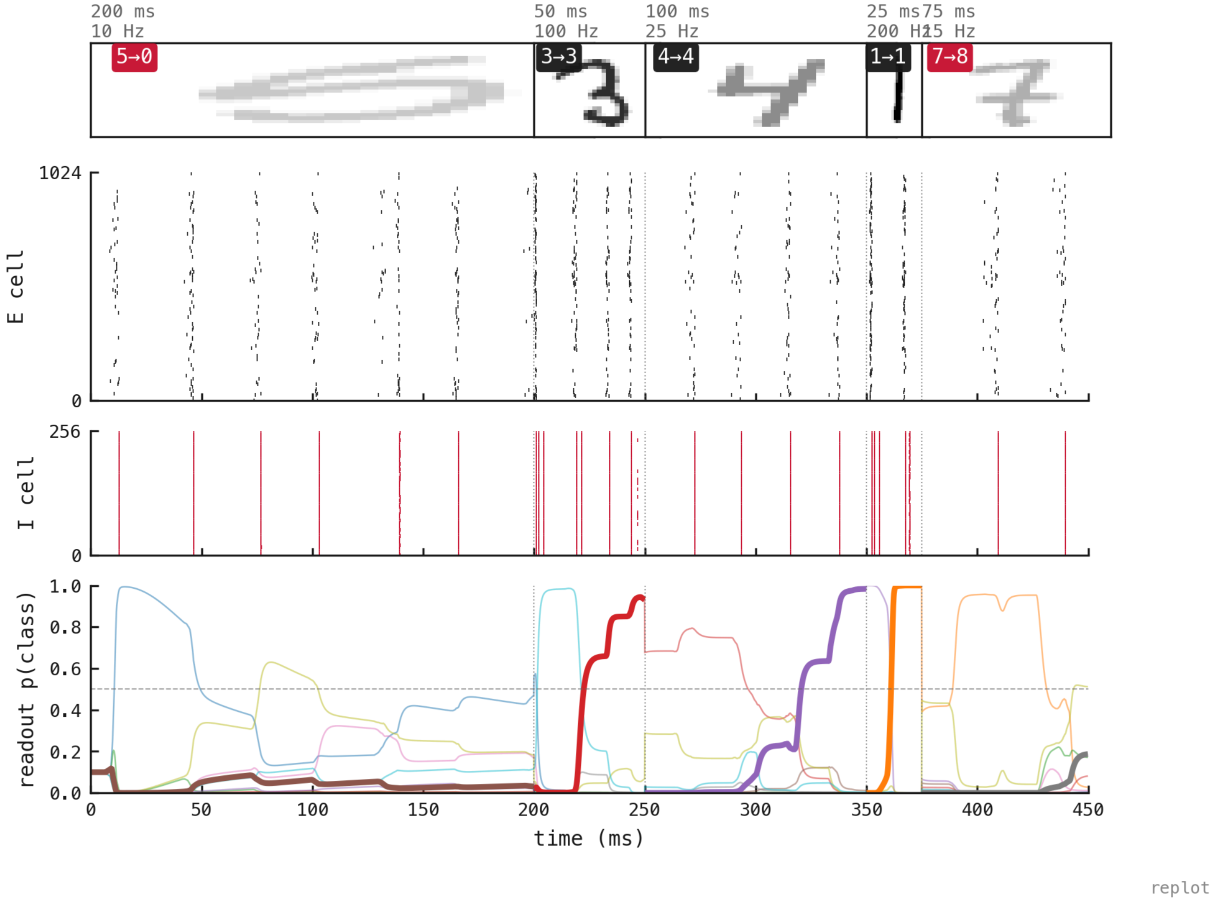

Figure 2: A harder test: each digit gets its *own* duration and input rate within one stream (durations 25–200 ms, rates 10–200 Hz). Thumbnail opacity ∝ input rate (faint = weak drive, bold = strong); the sliding window uses each segment's own τ, so every digit's prediction respects its presentation window. **3/5 correct** here: the two errors fall on the weakest-drive segments (a 5 at 10 Hz and a 7 at 15 Hz), the two lowest input rates in the stream.

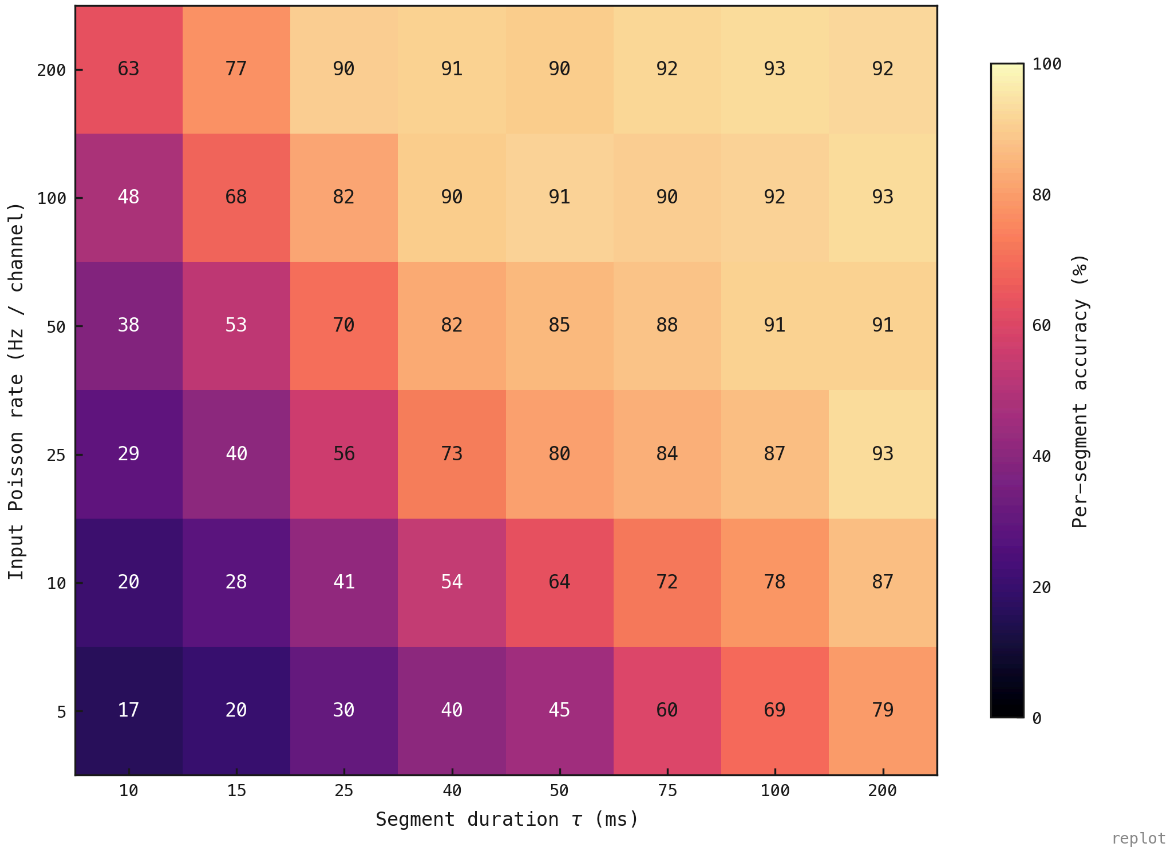

Figure 3: Per-segment accuracy over the full grid (3 seeds $\times$ 1200 segments per cell). Three things stand out. **(1) Sub-cycle failure below $\tau \approx$ 15 ms:** at τ = 10 ms even 200 Hz input reaches only 63%, so the network cannot classify within less than one cycle regardless of drive, the cleanest evidence that the **gamma cycle is the temporal quantum** of its classification ability. **(2) Above one cycle, accuracy $\approx$ f(τ · rate):** iso-accuracy contours run diagonal, so τ and input rate substitute for each other, more drive compensating for shorter presentation and vice versa. **(3) The trained operating point** (200 ms, 25 Hz) sits at $\approx$ 93%, essentially the plateau maximum ($\approx$ 93% grid-wide), so it is already near-optimal: extra τ or drive buys nothing.